Rahul Dange

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Sydney, Australia · Open to relocation · Full work rights (Australia)

Summary

Computer engineer focused on C/C++ systems programming and backend development, with experience delivering full-stack applications and leading project teams. Builds and experiments with memory management, concurrency, filesystems and language implementation. Academic Tutor and Master of Computer Science graduate at the University of Sydney.

Technical Skills

Languages: C, C++, JavaScript, Python, Java, SQL, x86-64 Assembly
Systems: POSIX threads, memory management, atomics, processes, IPC, Linux, GDB, Valgrind
Web & tools: React, Node.js, Express, REST APIs, PostgreSQL, MongoDB, Git, Make, Bash

Experience

Academic Tutor Jan 2026 - Present
University of Sydney · Part-time

- Teach C and systems programming, including memory management, pointers, processes, IPC and concurrency. Support students with debugging and reasoning about program behaviour using GDB and Valgrind.

System Engineer Jan 2026 - Aug 2026
Alethi · Part-time, project-based

- Contributed low-level software development and systems architecture to System One's MCK1 kernel/compiler initiative. Explored photonic integrated-circuit concepts and prototype models as preliminary research.

Project Lead & Full-Stack Developer Aug 2025 - Nov 2025
University of Sydney × Aeoflo · Industry capstone

- Built and delivered a GEO auditing application using React, Node.js and Chromium-based page extraction. Grouped 304 candidate parameters by functional behaviour and used LLM-based experiments to investigate their importance and inform scoring.

Full Stack Engineer Feb 2021 - Jul 2023
ChessVerst · Part-time

- Translated requirements into specifications, wireframes and platform architecture. Developed frontend interfaces, backend services, database functionality and APIs, working across UI/UX, testing and integration.

Associate Software Engineer Mar 2022 - Aug 2022
Lyte Calc Technologies Pvt Ltd · Internship

- Developed e-commerce functionality using React, Node.js, Express and MongoDB. Completed and handed over a reusable application template and participated in candidate interviews for the project team.

Projects

MIMFS | Multithreaded In-Memory File System | C

- Built a POSIX-style filesystem within caller-provided memory, without dynamic allocation. Supports directories, hard/symbolic links, sparse files and rename, with C11 atomic allocation and per-inode locks.

LLFPL | Low-Level Floating-Point Language | C

- Built a numerical expression interpreter that memory-maps source and evaluates without constructing a syntax tree. Reserves storage at startup to avoid heap allocation during evaluation; uses IEEE 754 double precision.

Education

University of Sydney Feb 2024 - Dec 2025
Master of Computer Science (Software Engineering) *Sydney, Australia*
Nutan College of Engineering and Research Aug 2019 - Aug 2023
Bachelor of Technology in Computer Science and Engineering *Pune, India*

Leadership & Volunteering

NCER Hackathon coordinator (2022), mentoring eight teams; Student Advisor (2022–2023); Cyber Security Club founder/coordinator (2019–2022).

Publication

Revolutionizing Student Engagement: Building a Cutting-Edge Interaction System. ICCIP 2023 / SSRN. DOI 10.2139/ssrn.4625799.